

Lecture 7 & 8 – Modeling & Design

50 MCQs (Dr.'s Exam Style)

1. Modeling helps us to:

- A) Eliminate reality
- B) Understand and change reality
- C) Replace implementation
- D) Avoid design

Answer: B

2. According to George Box:

- A) All models are correct
- B) All models are wrong, but some are useful
- C) Models replace code
- D) Models are perfect

Answer: B

3. Analysis modeling focuses on understanding:

- A) Internal structure
- B) Coding style
- C) The nature of the problem
- D) Server configuration

Answer: C

4. Design modeling focuses on understanding:

- A) User requirements
- B) Internal structure of the WebApp

- C) Business goals
- D) Testing strategy

Answer: B

5. A modeling language includes:

- A) Code only
- B) Notations, symbols, and rules
- C) Testing tools
- D) Servers

Answer: B

6. UML is considered a:

- A) Web-specific modeling language
- B) General-purpose modeling language
- C) Programming language
- D) Database language

Answer: B

7. UML structure diagrams include:

- A) Use Case Diagram
- B) Sequence Diagram
- C) Class Diagram
- D) Activity Diagram

Answer: C

8. UML behavior diagrams include:

- A) Class Diagram
- B) Component Diagram
- C) Use Case Diagram
- D) Deployment Diagram

Answer: C

9. UML interaction diagrams include:

- A) Activity Diagram
- B) State Diagram
- C) Sequence Diagram
- D) Package Diagram

Answer: C

10. WebML is designed mainly for:

- A) Transaction systems
- B) Data-intensive WebApps
- C) Mobile apps
- D) Embedded systems

Answer: B

11. The WebML structural model defines:

- A) Navigation links
- B) Data content using entities and relationships
- C) Page layout
- D) User groups

Answer: B

12. The WebML hypertext model consists of:

- A) Structural and presentation models
- B) Composition and navigation models
- C) Content and function models
- D) User and role models

Answer: B

13. Analysis modeling examines all EXCEPT:

- A) Content
- B) Interaction
- C) Behavior
- D) Source code

Answer: D

14. One factor that affects how much analysis modeling is done is:

- A) Programming language
- B) WebApp increment size
- C) UI color
- D) Server brand

Answer: B

15. Interaction model describes:

- A) Database structure
- B) User interaction with the WebApp
- C) Internal algorithms
- D) Network topology

Answer: B

16. Which is NOT an analysis output?

- A) Interaction model
- B) Information model
- C) Functional model
- D) Component model

Answer: D

17. Information model identifies:

- A) Processing logic
- B) Web server details
- C) Full spectrum of content
- D) Navigation rules

Answer: C

18. Functional model defines:

- A) Page layout
- B) User interface colors
- C) Operations applied to content
- D) Hardware environment

Answer: C

19. Configuration model describes:

- A) User behavior
- B) Database schema
- C) Environment and infrastructure
- D) Navigation paths

Answer: C

20. Use cases are primarily used to describe:

- A) Database queries
- B) Code execution
- C) User interaction scenarios
- D) Server setup

Answer: C

21. When interaction is complex, it is useful to use:

- A) Plain text only
- B) Rigorous diagrams
- C) Source code
- D) Database tables

Answer: B

22. A prototype supports:

- A) Coding
- B) Requirement validation with clients
- C) Deployment
- D) Performance tuning

Answer: B

23. Content objects may include:

- A) Only databases
- B) Only text
- C) Events and occurrences
- D) Only users

Answer: C

24. Data trees are used to depict:

- A) Navigation flow
- B) Content hierarchy
- C) System performance
- D) User roles

Answer: B

25. Activity diagrams illustrate:

- A) Static structure
- B) Processing flow and decisions

- C) Server architecture
- D) User roles

Answer: B

26. State diagrams are most useful when:

- A) Content changes
- B) User triggers state changes
- C) Data is static
- D) No interaction exists

Answer: B

27. Configuration modeling considers:

- A) Server hardware and OS
- B) Programming language syntax
- C) UI layout
- D) Database queries

Answer: A

28. Deployment diagrams are used when:

- A) Configuration is simple
- B) Architecture is complex
- C) UI is simple
- D) Only one server exists

Answer: B

29. Design should NOT be taken lightly when:

- A) WebApp is small
- B) Only one user exists
- C) WebApp impacts business success
- D) No database is used

Answer: C

30. Design model includes all EXCEPT:

- A) Content design
- B) Navigation design
- C) Component-level design
- D) Project scheduling

Answer: D

31. Simplicity in design means:

- A) Adding more features
- B) Feature-bloat
- C) Moderation
- D) Ignoring requirements

Answer: C

32. Consistency applies to:

- A) Content only
- B) Graphics only
- C) Navigation mechanisms
- D) All of the above

Answer: D

33. Navigability means:

- A) Fast servers
- B) Easy movement without searching
- C) Color consistency
- D) Database optimization

Answer: B

34. Compatibility requires WebApps to work across:

- A) One browser
- B) One OS
- C) Multiple environments
- D) One device

Answer: C

35. Quality requirements tree includes:

- A) Cost and profit
- B) Usability, functionality, reliability
- C) Coding speed
- D) Team size

Answer: B

36. Response time is related to:

- A) Usability
- B) Reliability
- C) Efficiency
- D) Maintainability

Answer: C

37. Ease of correction is related to:

- A) Reliability
- B) Maintainability
- C) Usability
- D) Efficiency

Answer: B

38. Interface design describes:

- A) Database design
- B) UI structure and navigation
- C) Server architecture
- D) Business rules

Answer: B

39. Aesthetic design focuses on:

- A) Workflow
- B) Look and feel
- C) Functional logic
- D) Data storage

Answer: B

40. Content design defines:

- A) UI colors
- B) Layout and structure of content
- C) Server processes
- D) User authentication

Answer: B

41. Navigation design describes:

- A) Data flow
- B) User movement between content
- C) Hardware interaction
- D) Database relations

Answer: B

42. Architecture design identifies:

- A) Page layout
- B) Overall WebApp structure
- C) User roles
- D) UI colors

Answer: B

43. Component design focuses on:

- A) User interaction
- B) Detailed processing logic
- C) Navigation links
- D) Business goals

Answer: B

44. Conceptual architecture should be developed:

- A) Per increment only
- B) After deployment
- C) In context of all increments
- D) Only for testing

Answer: C

45. Conceptual architecture represents:

- A) UI mockups
- B) Major functional and information components
- C) Code files
- D) Server IPs

Answer: B

46. Technical architecture shows:

- A) User scenarios
- B) Mapping to technical components

- C) Navigation flow
- D) Content hierarchy

Answer: B

47. Changing one technical component decision may:

- A) Affect nothing
- B) Affect other components
- C) Reduce risk automatically
- D) Improve performance always

Answer: B

48. Functional subsystems are identified from:

- A) UI design
- B) Key functionalities
- C) Database tables
- D) Deployment diagrams

Answer: B

49. ClientInterface subsystem provides:

- A) Database management
- B) Maintenance functions
- C) Interface between users and subsystems
- D) Security monitoring

Answer: C

50. SystemMaintenance subsystem handles:

- A) User authentication
- B) Order processing
- C) Database cleaning and maintenance
- D) UI navigation

Answer: C

Final Dr-Style Prediction (Lectures 7 & 8)

Expect **many questions** on:

- UML diagram classification
- Analysis vs Design modeling
- Design goals (simplicity, consistency, navigability)
- Architecture vs component design
- Quality attributes (usability, efficiency, maintainability)